

List data type

A collection of homogenous as well as heterogeneous objects enclosed within square brackets [] separated by commas is called a List. So we can say list is a sequential collection of elements.

1. Collection of objects enclosed within square brackets [] is called a list.
2. Duplicates are allowed.
3. Ordered / Sequential collection.
4. Indexing supported.
5. Slicing supported.
6. Different data types allowed.
7. Mutable in nature (elements can be modified).

Example:-

```
l1 = [10, 20, 30, 40]
l2 = ["Python", "Java", "C++"]
l3 = [10, "Python", 3.14, True]
```

```
print(l1)
print(l2)
print(l3)
```

Output

```
[10, 20, 30, 40]
['Python', 'Java', 'C++']
[10, 'Python', 3.14, True]
```

Accessing Elements (Indexing):-

```
l = [10, 20, 30, 40]
```

```
print(l[0])
print(l[2])
```

Output

```
10
30
```

Slicing:-

```
l = [10, 20, 30, 40, 50]
```

```
print(l[1:4])
```

Output

```
[20, 30, 40]
```

Python In-built functions for List:

Step	Syntax	Explanation
1	max()	To check maximum element in list
2	min()	To check minimum element in list
3	type()	To check datatype of object
4	id()	To check memory address of object
5	print()	To print output
6	len()	To count number of elements
7	list()	To create empty list or type casting
8	sum()	To calculate sum of list elements

Example:-

```
l = [10, 20, 30, 40]
```

```
print(max(l))
```

```
print(min(l))
```

```
print(type(l))
```

```
print(id(l))
```

```
print(len(l))
```

```
print(list())
```

```
print(sum(l))
```

Output

```
40
```

```
10
```

```
<class 'list'>
```

```
140404579813824
```

```
4
```

```
[]
```

```
100
```

List Methods

1. append()
2. extend()
3. insert()
4. remove()
5. pop()
6. clear()
7. index()
8. count()
9. sort()
10. reverse()
11. copy()

1. append():-- Add element at end:-

```
l = [10, 20, 30]
l.append(40)

print(l)

# O/P: [10, 20, 30, 40]
```

2. extend():-- Add multiple elements:-

```
l1 = [10, 20]
l2 = [30, 40]

l1.extend(l2)
print(l1)

# O/P: [10, 20, 30, 40]
```

3. insert():-- Insert element at specific index:-

```
l = [10, 20, 40]
l.insert(2, 30)
print(l)

# O/P: [10, 20, 30, 40]
```

4. remove():-- Remove specific element:-

```
l = [10, 20, 30, 40]
l.remove(20)
print(l)
```

```
# O/P: [10, 30, 40]
```

5. pop():-- Remove element using index number:-

```
l = [10, 20, 30]
l.pop()                # default it target last index
print(l)
```

```
# O/P: [10, 20]
```

6. clear():-- Remove all elements:-

```
l = [10, 20, 30]
l.clear()
print(l)
```

```
# O/P: []
```

7. index():-- Find index of element:-

```
l = [10, 20, 30, 40]
print(l.index(30))
```

```
# O/P: 2
```

8. count():-- Count occurrences:-

```
l = [10, 20, 10, 30, 10]
print(l.count(10))
```

```
# O/P: 3
```

9. sort():-- Sort elements:-

```
l = [40, 10, 30, 20]
l.sort()
print(l)
```

```
# O/P: [10, 20, 30, 40]
```

10. reverse():-- Reverse list:-

```
l = [10, 20, 30]
l.reverse()
print(l)
```

```
# O/P: [30, 20, 10]
```

11. copy():-- Copy list:-

```
l1 = [10, 20, 30]  
l2 = l1.copy()  
print(l2)
```

```
# O/P: [10, 20, 30]
```